

**The Impact of Workplace Culture and Gender Bias on Women's Retention in
Cybersecurity**

Taylor Myers

University of Maryland Global Campus

Writing 112

Dr. Barb Rowland

June 16, 2026

The Impact of Workplace Culture and Gender Bias on Women's Retention in Cybersecurity

More than half of women in information technology leave the field after only five years, creating a turnover rate twice as high as men in the same field (Beveridge, 2021). This statistic shows that the problem is not only getting women into cybersecurity but also keeping them in the field once they are there. Many women face workplace barriers such as isolation, stereotypes, limited mentorship, and male-dominated workplace cultures that can make them feel as though they do not fully belong. These issues can affect whether women feel supported enough to stay and grow in their careers. A diverse cybersecurity workforce is also important because different experiences can lead to stronger problem-solving and more creative responses to cyber threats. Osman et al. (2023) explains that a lack of diversity can limit the field's ability to recognize and respond to different forms of cyberattacks. Therefore, retaining women is not only an issue of fairness; it is also necessary for improving the effectiveness and resilience of the cybersecurity workforce. **Retention means more than hiring women into cybersecurity roles; it also means creating workplace conditions that encourage them to stay, advance, and build long-term careers. Without that support, organizations may continue losing qualified women even while the demand for cybersecurity professionals grows.** Workplace culture and gender bias negatively affect the retention of women in cybersecurity by creating barriers to belonging, limiting mentorship and advancement opportunities, and reinforcing stereotypes about who fits in this field, beginning with workplace environments that often make women feel excluded.

Workplace Culture and Barriers to Belonging

First, workplace culture creates barriers to belonging that negatively affect the retention of women in cybersecurity because women may feel isolated in male-dominated environments that are not always inclusive. Beveridge (2021) explains that women in information technology often leave because of barriers such as stereotypes, isolation, questions about legitimacy, masculine workplace climates, and work-life balance challenges. This shows that the issue is not only about whether women are qualified for cybersecurity roles, but whether the workplace environments make them feel accepted and supported enough to stay. When women feel like they constantly must prove they belong, it can make the field feel discouraging instead of welcoming.

Workplace culture can also affect women before they can enter the cybersecurity workforce. Shan and Yang (2022) explain that negative stereotypes and the “masculinization” of technology can make multiple women feel like they do not fit the image of a cybersecurity professional. To examine how early cybersecurity opportunities affect women’s long-term engagement, Shan and Yang (2022) surveyed fifty-five members of the Aspirations in Computing community and analyzed their responses using quantitative methods. The study found that participating in one cybersecurity program did not significantly affect long-term engagement, but repeated programs and greater access to cybersecurity resources increased girls’ interest in the field (Shan & Yang, 2022). The results of Shan and Yang’s (2022) study also show why belonging must be developed over time. Repeated exposure gives girls more chances to build skills, gain confidence, and view cybersecurity as a realistic career path. When organizations continue that support through internships, onboarding, and early-career development, women

may be more likely to remain engaged instead of feeling unprepared or excluded when entering the workforce. This matters because stereotypes about who belongs in technology can continue from school into the workplace. If women already feel underrepresented before entering the field, then entering a male-dominated workplace can make that feeling even stronger. As a result, workplace culture can create both professional and emotional barriers that make it harder for women to see themselves staying in cybersecurity long-term.

In addition, Bagchi-Sen et al. (2010) explain that information technology has often been connected to “hacker culture,” which can include long hours, late nights, and male-oriented expectations. The researchers conducted one of the first studies focused specifically on female cybersecurity professionals by interviewing women who had advanced into chief security officer and chief information officer positions (Bagchi-Sen et al., 2010). Their study examined the barriers women faced, the skills needed for cybersecurity careers, and how women defined professional success (Bagchi-Sen et al., 2010). **This research is important because it shows that even women who reach high-level cybersecurity positions still must deal with workplace cultures that were not always built with them in mind. If women must adjust to a culture that expects late nights, isolation, and male-centered networking, they may feel pressure to prove they belong instead of feeling fully accepted as professionals.** This type of environment can isolate women and make cybersecurity feel less inclusive. Radu and Smaili (2021) also show that women remain underrepresented in cybersecurity leadership, with women making up only 14% of the cybersecurity workforce in the United States. **This lack of representation matters because leadership affects workplace culture. When there are fewer women in leadership, there may also be fewer people in decision-making roles who understand the barriers women face or who can help create more inclusive policies,**

mentorship opportunities, and advancement paths. This lack of representation can make it harder for women to find mentors, role models, and leaders who understand their experiences. These feelings of isolation not only weaken women's sense of belonging but can also allow gender bias to shape their access to mentorship, professional support, and advancement opportunities.

Limiting Mentorship and Advancement Opportunities

Beyond workplace culture, gender bias limits mentorship and advancement opportunities for women in cybersecurity by creating unequal expectations and making it harder for women to feel respected and supported in their roles. Women may have the education, skills and experience needed to succeed, but stereotypes can still cause employers and coworkers to question their abilities. Beveridge (2021) explains that women in technology often face questions about their legitimacy and assumptions about whether they are qualified for technical roles. When women must repeatedly prove that they are capable, they may feel undervalued and receive fewer opportunities to take on important responsibilities that could help advance their careers.

Mentorship is another important part of career growth because mentors provide guidance, encouragement, professional connections, and advice about leadership opportunities. However, women in cybersecurity do not always have equal access to these relationships. Harrison (2020) reported that only 7% of women in cybersecurity had been mentored by another woman, while 25% had never received mentorship at all. These findings came from the Women in Cybersecurity Survey (2010), which examines women's experiences with leadership, career advancement, certifications, professional visibility, and mentorship. The limited number of women in leadership roles may make it especially difficult for women to find mentors who

understand the challenges they experience. Without strong mentorship and professional support, women may have fewer opportunities to build confidence, expand their networks, and prepare for advancement. As cybersecurity professional Heather Mahalik explains, “Your title and time in the field do not define what you know and the impact you can provide to this community” (Harrison, 2020, para. 2). This statement shows why mentors and employers should recognize women’s knowledge and abilities instead of allowing titles, stereotypes, or limited experience to determine their potential.

Gender bias can also directly affect whether women receive promotions and leadership opportunities. Bagchi-Sen et al. (2010) found that women in cybersecurity face institutional barriers, stereotypes, and difficulties developing the professional networks needed to advance into executive positions. These barriers can become even greater for women who also belong to racial minority groups. Janofsky (2018) reported findings from a survey of approximately 9,500 cybersecurity professionals in the United States. ~~The survey~~ **Janorsky (2018)** found that 35% of Black women in cybersecurity experienced an unexplained denial or delay in career advancement, compared with only 4% of white women. ~~The study~~ **Janorsky (2018)** also found that minority cybersecurity professionals were 12% more likely to hold a master’s degree or higher but were 7% less likely to hold leadership positions. Diversity consultant Joelle Emerson emphasizes the importance of representation by stating, “If you’re trying to build products for a diverse world, it matters who is on your team” (as cited in Janofsky, 2018, para. 7). This quote reinforces that diverse teams are not only important for fairness but also for creating stronger cybersecurity solutions that reflect the needs of different people and communities. The differences reported by Janofsky (2018) also demonstrate how gender bias can overlap with racial bias. Black women may face barriers that are not fully explained by gender alone, even

when they have advanced education and professional qualifications. When organizations fail to examine how promotion decisions are made, unequal treatment can continue without being openly recognized. Clear promotion standards, documented evaluations, and accountability for leadership decisions can help reduce these hidden barriers. These findings show that unequal treatment can have a serious effect on women's career growth and retention. These barriers can cause qualified women to leave cybersecurity, organizations must respond by providing stronger mentorship, meaningful representation, and consistent workplace support.

Reinforcing Stereotypes About Who Fits in Cybersecurity

Workplace culture and gender bias also negatively affect the retention of women in cybersecurity by reinforcing stereotypes about who fits in the field. These stereotypes can make women feel as though they must prove they belong in technical and leadership roles before they are fully respected. To reduce these barriers, organizations must challenge ~~stereotypes about~~ **narrow ideas about who fits in** can succeed in cybersecurity by increasing mentorship, representation, and workplace support. Women need more than **an** opportunity to enter the field; they also need access to resources and professional relationships that help them remain involved and grow throughout their careers. Beveridge (2021) explains that organizations should create inclusive policies and environments that do not alienate women. Employers can help challenge gender stereotypes by recognizing women's contributions, encouraging them to pursue technical and leadership roles, and ensuring they receive the same opportunities as their coworkers.

Osman et al. (2023) provides additional evidence that mentorship and workplace support can help underrepresented individuals remain engaged in cybersecurity. The researchers

conducted structured interviews and surveys with twenty-three participants, including underrepresented cybersecurity professionals, college students, instructors, and a high-school administrator (Osman et al., 2023). Through a thematic analysis, they identified seventeen themes related to entering the field, overcoming barriers, developing skills, and maintaining engagement (Osman et al., 2023). Their findings showed that mentorship, representation, social support, and a sense of belonging can help women and other underrepresented individuals persist in cybersecurity (Osman et al., 2023). **This matters because retention depends on more than individual motivation; it also depends on whether women have support systems that help them feel included, prepared, and encouraged to keep growing in the field.**

Providing clear professional-development opportunities can also help women prepare for advancement instead of leaving their career growth to chance. Harrison (2020) reported that 41% of surveyed women credited being in the right place at the right time for their rise into leadership positions, while 38% credited gaining varied professional experiences and 34% credited pursuing certifications. Although women can take steps to strengthen their qualifications, employers should also provide structured opportunities that allow them to gain experience and become visible to decision-makers. Leadership training, certification support, and opportunities to participate in important projects can show women their organization values their abilities and wants them to succeed.

Meaningful representation can also improve organizational cybersecurity decisions. Radu and Smaili (2021) examined three hundred company-year observations from organizations listed on the S&P/TSX 60 Index between 2014 and 2018. The researchers analyzed corporate reports to determine whether board gender diversity affected cybersecurity-related disclosures (Radu &

Smaili, 2021). They found that women's representation had a positive effect on cybersecurity disclosure, but the strongest results occurred when boards included at least three women (Radu & Smaili, 2021). This study shows that representation must be meaningful enough for women to influence leadership decisions rather than being included only symbolically. Radu and Smaili's (2021) findings also demonstrate why representation should extend beyond hiring women into entry-level positions. When women are included in leadership in meaningful numbers, they have greater opportunity to influence organizational priorities and cybersecurity decisions. Visible representation can also show women earlier in their careers that advancement is possible, which may strengthen their motivation to remain with an organization and pursue leadership roles.

Programs that consistently connect women with education, employers, mentors, and other cybersecurity professionals can provide important long-term support. Women in CyberSecurity offers resources such as mentoring programs, professional affiliates, student chapters, job boards, career fairs, mock interviews, and resume reviews (Women in CyberSecurity, 2021). Osman et al. (2023) similarly found that mentorship provides professional guidance, promotes a sense of belonging, and helps women persist and seek advancement in cybersecurity. The researchers recommend creating formal mentorship programs and connecting underrepresented individuals with cybersecurity professionals who share similar backgrounds and experiences. Shan and Yang (2022) also found that offering cybersecurity programs only once was not enough to increase girls' long-term interest; repeated opportunities and greater access to resources were more effective. Together, these findings show that continued mentorship and professional support are more valuable than one-time diversity efforts. These continued efforts are essential because improving women's retention requires organizations to move beyond recruitment and create environments where women can remain, advance, and succeed.

Conclusion

Overall, improving the retention of women in cybersecurity requires organizations to look beyond recruitment and address the workplace conditions that affect whether women choose to stay. Male-dominated environments, feelings of isolation, and lack of inclusion can make women feel as though they do not fully belong in the field. Gender bias can also limit access to mentorship, professional networks, leadership opportunities, and fair advancement, even when women have the skills and qualifications needed to succeed. These challenges can reinforce harmful stereotypes about who fits in cybersecurity and makes long-term career growth more difficult.

Organizations can help address these problems by creating more inclusive workplace cultures, providing structured mentorship opportunities, and giving women clear paths toward professional growth and leadership. Consistent support, meaningful representation, and equal access to training and advancement can help women feel respected and valued in their roles. Retaining more women would not only improve fairness within the cybersecurity workforce but would also strengthen the field by bringing in a wider range of experiences, ideas, and problem-solving approaches. For cybersecurity organizations to meet growing workforce demands, they must create environments where women are not only encouraged to enter the field but are also supported enough to remain and succeed. Organizations should regularly review retention, promotion, mentorship participation, and employee feedback to determine whether their inclusion efforts are working. Simply creating a diversity policy is not enough if women continue to feel isolated or overlooked for advancement. Lasting improvement will require

leaders to listen to women's experiences, measure organizational progress, and correct workplace practices that continue to create unequal outcomes.

Resources

Bagchi-Sen, S., Rao, H. R., Upadhyaya, S. J., & Chai, S. (Volume: 12, Issue: 1, Jan.-Feb. 2010). (PDF) *Women in cybersecurity: a study of career advancement*. IEEE Xplore logo
- Link to home. <https://ieeexplore.ieee.org/document/5403174>

Beveridge, R. (2021). *Addressing the gender gap in the cybersecurity workforce*. *International Journal of Cyber Research and Education*, 3(2), 54–61.
<https://doi.org/10.4018/IJCRE.2021070105>

Chaparro Osman, Dr. M., Namukasa, M., Ficke, C., Piasecki, I., OConnor, Dr. T., & Carroll, Dr. M. (2023, October). *Understanding how to diversify the cybersecurity workforce: A qualitative analysis*. DigitalCommons@Kennesaw State University.
<https://digitalcommons.kennesaw.edu/jcerp/vol2023/iss2/4/>

Harrison, L. (2020, March). *Leadership roles of women in cybersecurity: women control their own leadership destiny; mentorship counts*. PR Newswire.
<http://ezproxy.umgc.edu/login?url=https://www.proquest.com/wire-feeds/leadership-roles-women-cybersecurity/docview/2369377333/se-2?accountid=14580>

Janofsky, A. (2018). *Nearly one-third of minority cybersecurity professionals in U.S. report workplace discrimination*. *WSJ Pro.Cyber Security*, <http://ezproxy.umgc.edu/login?url=https://www.proquest.com/trade-journals/nearly-one-third-minority-cybersecurity/docview/2171081491/se-2>

- Radu, C., & Smaili, N. (2022). *Board gender diversity and corporate response to cyber risk: Evidence from cybersecurity related disclosure*. *Journal of Business Ethics*, 177(2), 351–374. <https://doi.org/10.1007/s10551-020-04717-9>
- Shan, M., & Yang, J. (2022). *Investigating the accessibility and impacts of cybersecurity programs on high-school girls' long-term industry engagement*. *Information and Computer Security*, 30(3), 309-323. . <https://doi.org/10.1108/ICS-05-2021-0067>
- Women in Cybersecurity (wiCyS). (2021, October). *Sponsorship opportunities available for 2022 women in cybersecurity (WiCyS) conference*. PR Newswire.
<http://ezproxy.umgc.edu/login?url=https://www.proquest.com/wire-feeds/sponsorship-opportunities-available-2022-women/docview/2583941942/se-2?accountid=14580>